

GREENVILLE DOWNTOWN, SC, USA

WMO: 723119

Lat: 34.846N

Lon: 82.346W

Elev: 1048

StdP: 14.15

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 13886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	26.8	1.2	6.1	30.8	5.3	7.5	33.5	22.2	44.1	20.0	47.6	6.1	10	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	94.0	73.0	92.0	72.9	90.1	72.4	76.5	87.4	75.5	85.7	74.8	84.7	7.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	129.1	79.5	72.7	125.8	79.2	72.1	123.3	78.9	40.4	87.9	39.6	86.5	38.9	84.7	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.9	15.2	DB	16.4	97.3	3.8	3.9	13.7	100.1	11.4	102.4	9.3	104.6	6.5	107.4
			WB	13.3	77.7	3.5	0.9	10.8	78.4	8.8	78.9	6.8	79.4	4.3	80.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	43.1	46.8	54.2	62.7	70.4	77.5	80.0	79.4	73.9	63.2	53.2	46.2
	DBStd	14.88	9.39	8.68	8.77	7.44	6.23	4.31	3.88	3.93	5.75	7.41	7.82	8.27
	HDD50	705	256	151	58	6	0	0	0	0	0	4	55	175
	HDD65	2762	680	511	346	128	27	1	0	0	7	120	359	583
	CDD50	5322	42	62	190	386	634	826	931	912	717	414	151	57
	CDD65	1903	0	2	13	59	196	376	466	447	273	65	5	1
	CDH74	16979	2	6	103	529	1562	3419	4585	4154	2137	451	29	2
	CDH80	6605	0	0	14	107	474	1396	2016	1735	769	93	1	0
(14) Wind	WSAvg	6.6	6.8	7.1	7.6	7.5	6.9	6.3	6.0	5.7	6.5	6.3	6.1	6.2
(15) Precipitation	PrecAvg	50.5	4.2	3.9	4.8	3.9	3.3	4.2	4.9	4.9	4.4	3.3	3.5	4.8
	PrecMax	66.7	7.5	7.0	8.9	9.4	7.6	9.5	11.7	15.6	13.4	8.4	8.6	9.8
	PrecMin	34.1	1.6	1.7	1.6	1.6	1.1	0.3	2.3	0.7	0.2	0.0	0.8	2.3
	PrecStd	9.4	1.6	1.2	1.9	1.8	1.6	2.4	2.4	3.0	2.6	2.2	1.7	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.8	74.2	82.1	86.5	91.8	95.3	97.1	97.4	93.6	87.3	78.5	71.8
		MCWB	57.3	61.2	61.4	65.3	70.0	72.3	73.7	72.4	71.3	71.5	63.2	60.6
	2%	DB	65.6	69.6	76.7	83.0	88.0	92.6	93.9	93.3	90.4	82.0	73.1	67.3
		MCWB	56.6	57.2	59.1	64.4	68.7	72.2	74.0	72.9	70.5	66.8	60.1	59.4
	5%	DB	62.3	65.6	72.8	80.2	85.1	90.3	91.6	90.9	87.7	79.0	69.9	63.5
		MCWB	54.0	55.2	57.7	63.0	68.2	71.6	74.0	73.2	70.4	65.0	59.1	55.8
	10%	DB	57.6	61.5	69.2	76.7	82.1	87.8	89.7	88.5	84.5	75.5	66.2	59.5
		MCWB	50.8	51.8	56.1	60.7	66.9	71.1	73.4	72.9	69.4	64.2	57.1	53.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.2	65.1	64.8	70.1	74.1	76.3	77.8	77.5	75.5	74.3	68.3	65.4
		MCDB	66.9	71.3	73.8	80.9	82.6	88.4	90.0	88.5	85.0	81.1	71.0	68.3
	2%	WB	59.5	61.7	62.9	67.9	72.1	75.0	76.7	76.4	74.4	71.4	64.7	61.6
		MCDB	63.7	66.1	71.2	77.8	81.2	86.6	88.2	86.6	83.1	78.0	68.7	65.8
	5%	WB	55.7	57.9	60.9	65.9	70.7	73.9	75.7	75.4	73.2	68.6	62.4	58.3
		MCDB	59.7	63.0	69.3	74.9	80.0	84.9	86.4	84.9	81.1	75.2	66.7	61.8
	10%	WB	52.3	53.6	58.7	63.8	69.2	72.9	74.9	74.6	71.9	66.3	59.4	54.2
		MCDB	56.1	59.1	66.2	72.3	78.9	83.3	85.2	83.8	79.7	72.7	64.9	57.9
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	18.6	19.9	20.7	18.9	17.9	17.2	16.7	16.9	18.7	19.4	16.9
		MCDBR	21.7	22.7	23.7	24.2	21.9	20.9	19.8	19.8	20.3	21.7	22.7	20.0
	5% WB	MCWBR	14.0	13.9	11.8	10.2	8.0	6.6	6.0	6.0	6.3	9.3	12.2	13.0
		MCWBR	17.3	18.9	20.2	20.3	18.2	17.6	17.5	16.6	15.9	16.2	16.3	15.6
(40) Clear-Sky Solar Irradiance	taub		0.309	0.319	0.353	0.392	0.450	0.476	0.521	0.510	0.414	0.377	0.332	0.306
		taud	2.512	2.483	2.415	2.305	2.175	2.139	2.038	2.053	2.305	2.389	2.474	2.549
	Ebn at Noon		286	295	292	284	267	259	246	246	268	269	273	280
		Edh at Noon	26	30	35	41	47	49	54	52	39	33	27	24
(44) All-Sky Solar Radiation	RadAvg		845	1046	1385	1727	1908	2011	1885	1742	1498	1246	939	727
	RadStd		48	118	92	126	147	142	122	83	133	141	84	70

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.80	N/A	N/A	N/A	N/A	N/A	N/A	+527	+344
(47) Regional (49 neighbors)		+0.99	N/A	N/A	N/A	N/A	N/A	N/A	+304	+207

Nomenclature: See separate page